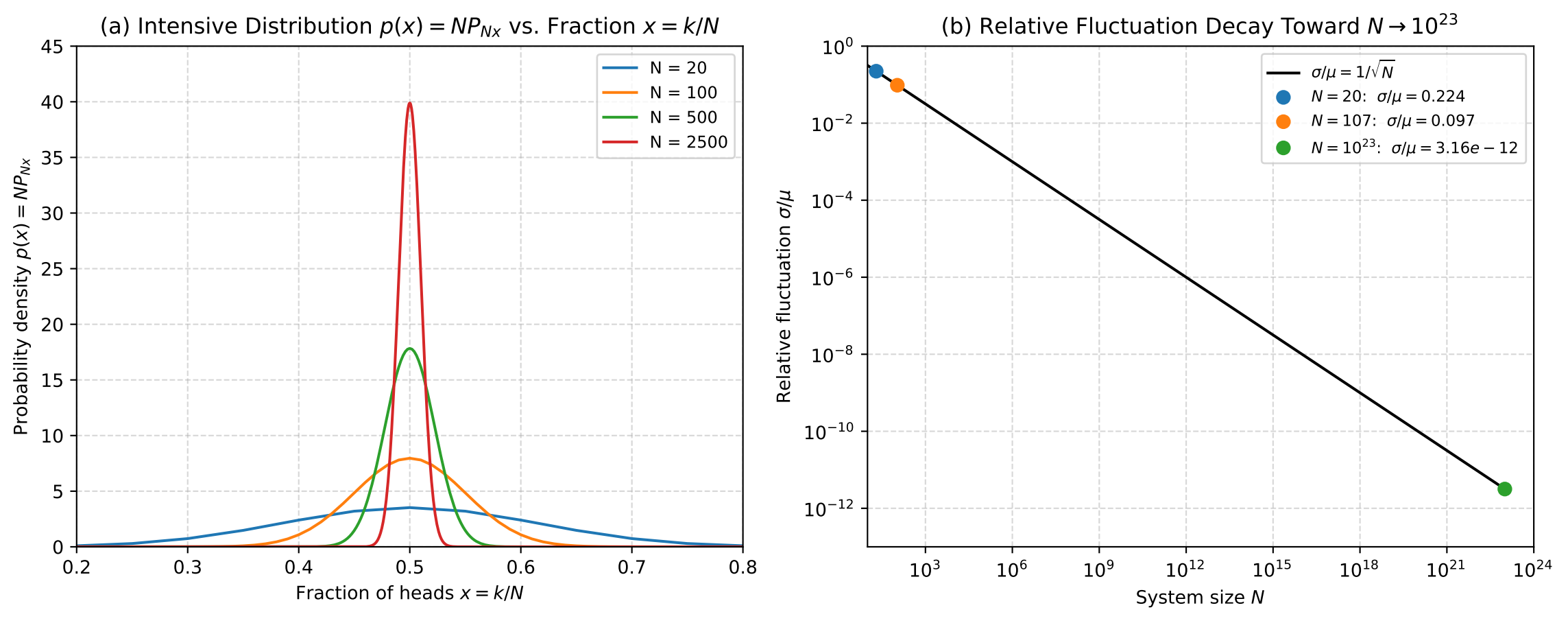
(a) Intensive Distribution $p(x) = NP_{Nx}$ vs. Fraction $x = k/N$
Probability density $p(x) = NP_{Nx}$
Fraction of heads $x = k/N$
N = 20
N = 100
N = 500
N = 2500
(b) Relative Fluctuation Decay Toward $N \to 10^{23}$
Relative fluctuation σ/μ
System size N
$\sigma/\mu = 1/\sqrt{N}$
$N = 20$: $\sigma/\mu = 0.224$
$N = 107$: $\sigma/\mu = 0.097$
$N = 10^{23}$: $\sigma/\mu = 3.16e-12$